30. An exact finite genotype–phenotype map and its mutation graph

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Different sequences of genetic letters can produce the same modeled structure. That creates paths through sequence space along which mutations leave the chosen outcome unchanged. This problem constructs a small RNA-inspired map exactly and asks how those paths connect, how often one mutation changes the outcome, and which new structures are nearby. The aim is to study a controlled model of robustness and variation.

Key terms

- **Genotype** — The input sequence, here a string over the four RNA letters A, C, G and U.
- **Phenotype** — The selected output of the model, here a secondary-structure pattern rather than an organism's full observable traits.
- **Secondary structure** — A pattern of paired positions along an RNA sequence, with the allowed pairings and restrictions stated explicitly.
- **Nussinov-style model** — A combinatorial folding model maximizing the number of allowed pairs, not a full free-energy calculation.
- **Tie-breaking rule** — A deterministic rule choosing one output when several structures achieve the same score.
- **Hamming graph** — A graph whose vertices are sequences and whose edges join sequences differing at exactly one position.
- **Neutral component** — A connected group of sequences with the same modeled phenotype.
- **Mutational robustness** — The fraction of a genotype's one-letter changes that preserve its phenotype under the chosen map.
- **Evolvability** — Here, a specified count of different phenotypes accessible by mutation, not a complete measure of biological adaptability.
- **Fitness** — A separately defined measure of reproductive success or a model score; it is not determined automatically by the folding map.

Why this matters

The structure of a genotype–phenotype map helps explain how variation can accumulate without immediately changing a selected outcome, and how new outcomes become accessible. Exact small maps provide benchmarks for sampling methods and expose how conclusions depend on folding rules and tie choices.

What progress would mean

A complete short-sequence graph would establish precise robustness and connectivity facts for the model. Those facts could motivate biological hypotheses, but would not by themselves predict RNA function, population evolution or a universal transition at some sequence length.

Core question

For a fully specified combinatorial RNA folding rule on short sequences, what are the exact neutral components, mutation robustness, and accessible phenotype counts?

Scope and assumptions

Use alphabet {A,C,G,U}, sequence length L=4,…,8 initially, allowed pairs AU/UA, GC/CG and GU/UG, minimum hairpin length three unpaired bases, no pseudoknots, and one pair per base. Maximize pair count and break ties by lexicographically smallest dot-bracket string under a declared character order.

This defines a mathematical GP map, not a thermodynamic RNA predictor. A nearest-neighbor free-energy extension must name its parameter set; empirical parameters are then acknowledged model inputs.

Mathematical target

Let Φ map each sequence to its deterministic selected structure. On the Hamming graph H(L,4), exact-one-mutation neighbors satisfy d_H(s,t)=1 and each sequence has 3L neighbors. Define

$$r(s) = \frac{|\{t : d_H(s,t) = 1,\ \Phi(t) = \Phi(s)\}|}{3L}.$$

For each phenotype analyze the induced neutral graph and every connected component; a phenotype preimage need not be connected. Define evolvability as the number of distinct other phenotypes reachable by one mutation, stating whether it is measured per genotype, component or whole preimage.

Required outputs and proof obligations

Export folding rules, tie-breaking, sequence-to-structure table, neutral components, exact counts and rational robustness. A search restricted to a component must not report the full phenotype frequency. Fitness is a separately supplied function; evolutionary dynamics require population size, mutation kernel, selection rule and initial state.

Validation and rejection controls

Cross-check dynamic programming with exhaustive legal-structure enumeration on short sequences, especially ties and prohibited pairs. Verify $\Sigma_\phi|\Phi^{-1}(\phi)|=4^L$ and every Hamming vertex degree is 3L. Count neutral edges independently and recover average robustness from twice the edge count. Treat any larger-L sample as statistical evidence.

Milestones and research extension

Baseline: complete exact maps through L=8. Strong: increase L within measured resources and characterize component-level robustness/evolvability. Research extension: a proved asymptotic property or a specified noisy GP-channel capacity problem. Do not hard-code a universal percolation threshold or empirical target correlation.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- ViennaRNA, RNAfold documentation — RNAfold's energy model and parameter choices differ from a pair-count model.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\Phi : \{A,C,G,U\}^L \longrightarrow \mathcal{S}_L, \qquad r(s) = \frac{|\{t : d_H(s,t) = 1,\ \Phi(t) = \Phi(s)\}|}{3L},$$

$$\sum_{\varphi} |\Phi^{-1}(\varphi)| = 4^L, \qquad \deg_{H(L,4)} s = 3L.$$

Specify pair-count maximization, allowed pairs, hairpin length, pseudoknot exclusion and deterministic tie-breaking. Enumerate every connected component of each phenotype preimage. This combinatorial map is not a thermodynamic predictor. At $L = 20$, $4^L = 1{,}099{,}511{,}627{,}776$; sampling does not establish a complete catalog.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — two folding objectives (16, 61–71, 1083).** Nussinov pair maximization is not nearest-neighbor MFE minimization. Requiring 100% agreement with ViennaRNA's default energy model is an invalid acceptance test.
2. **Major — nondeterministic ties.** Without an explicit tie-breaking rule, the map is set-valued and neutral-network statistics depend on the chosen representative. The rewrite fixes the map before measuring it.
3. **Major — neighborhood definition (55, 83).** B_1 includes the genotype itself, so the original normalization is not the fraction of one-mutant neighbors. The correct degree is 3L.
4. **Major — feasibility and extrapolation (39, 87, 1102–1105).** 4^{20}=1,099,511,627,776 sequences; exhaustive folding is not a casual short-sequence baseline. A universal L≈30 percolation threshold and fixed robustness correlations do not follow from the model. Start with a complete tractable graph and report sampling separately.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a fully specified combinatorial RNA folding rule on short sequences, what are the exact neutral components, mutation robustness, and accessible phenotype counts?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: complete exact maps through L=8. Strong: increase L within measured resources and characterize component-level robustness/evolvability. Research extension: a proved asymptotic property or a specified noisy GP-channel capacity problem. Do not hard-code a universal percolation threshold or empirical target correlation.

Sources supporting the corrections

- ViennaRNA, RNAfold documentation — RNAfold's energy model and parameter choices differ from a pair-count model.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.